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Using data analytics to distinguish legitimate and illegitimate shell companies

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A B S T R A C T

Shell companies can be a legitimate entity but can also been used for illicit activities such as money laundering. Users of shell companies have included illegal arms dealers, drug cartels, terrorists and cyber-criminals, as well as legitimate businesses. To assist in distinguishing between legitimate and illegitimate uses of shell companies, we develop a data-driven model to detect shell companies that are being used for money laundering. We use a hybrid approach combining graph analytics and supervised machine learning. The resulting detection models have an impressive classification accuracy ranging between 88.17 % and 97.85 %. We found no prior study that developed such models to detect illicit shell companies using publicly available information as done with our models. Beneficiaries of this work include government officials and compliance professionals, particularly accountants, tax officials and anti-corruption agencies.

1. Introduction

Money laundering, the process of giving dirty money a legitimate appearance, has been the subject of extensive debate among academics, practitioners, and regulators. The International Monetary Fund (IMF) estimated the aggregate level of money laundering to be between 2 % and 5 % of the world's annual gross domestic product or approximately 1.5 trillion US dollars (FATF, 2012). Within Australia, this figure equates to AUD 10–15 billion per annum (Singh and Best, 2019). These activities constitute a profound threat to the global economy, as they enable criminal enterprises, undermine the integrity of the financial system, and foster a shadow economy (Lyman, 2011; Schneider, 2010; Schneider and Windischbauer, 2008). Apart from a range of underground activities, such as drug trafficking, cybercrime, and corruption, money laundering is also linked to quasi-legal activities involving the concealment of income from public authorities, often facilitated by suspicious accounting practices (Ravenda et al., 2018). Furthermore, amongst the various mechanisms employed to undertake money laundering, shell companies play a critical role in hiding beneficial ownership and concealing financial flows.

These entities, while having legitimate uses, have also been manipulated as tools to launder money and hide information about beneficial owners (Gilmour, 2020). Notably, illicit arms dealers, drug cartels, corrupt politicians, terrorists, and cyber-criminals have emerged as users of shell companies. The academic literature has also examined the use of shell companies to avoid and evade taxes (Compin, 2008; Sikka and Willmott, 2010). The ease of setting up such entities, the transnationality involved, and the lack of adherence to Financial Action Task Force (FATF) guidelines have posed a challenge for law enforcement authorities in countering crime and corruption (Martini et al., 2019).

From a network perspective, money laundering exploits a web of disguised relationships, and unraveling this complex network of connections is suited to graph analytics. It can facilitate analysis of hidden networks, enabling investigators to detect patterns, infer relationships, and identify suspicious activities (Bangcharoensap et al., 2015; Battaglia et al., 2018; Liu et al., 2016; Morselli and Roy, 2008; Raghavan et al., 2007; Soramäki and Cook, 2013; Van Vlasselaer et al., 2017). Consequently, there has been a focus on using graph networks to develop detection models related to financial fraud. However, an obstacle while developing a detection model is the lack of data associated

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with a particular type of fraud. This leads to the creation of synthetic data that match closely to the actual data. An anomaly to this is the work of Ravenda, Argilés-Bosch, and Valencia-Silva (Ravenda et al., 2015), who used real-life data of illicit mafia firms in their analysis. Furthermore, previous works to detect illicit activities like money laundering have focused on banking transactions, which may not be publicly available and may increase the burden of tracking an audit trail (Luna et al., 2018; Savage et al., 2016; Singh and Best, 2019).

This paper builds on the work from Tiwari (2021) and Tiwari et al. (2022) to utilize graph database platforms in combatting money laundering. Specifically, this paper aims to detect money laundering activities by leveraging graph analytics and non-parametric supervised learning with publicly available data on shell companies identified in corruption cases. To achieve this, the present work addresses two key research questions: (1) How can graph analytics be leveraged to combat money laundering activities involving shell companies using publicly available data; (2) and what combinations of graph algorithms and supervised learning techniques provide the highest accuracy in distinguishing legitimate and illicit shell companies.

By addressing these research questions, this work establishes a framework for analysing publicly available data, with a focus on the structural and relational attributes of entities involved in money laundering, and provide actionable insights for key stakeholders, such as anti-corruption NGOs, corporate registries, financial institutions, government officials, and legal and compliance professionals, especially tax officials and accountants. For example, accountants play a vital role in reporting money laundering activities (Mitchell et al., 1998) and the proposed detection models can enhance their efforts. Similarly, financial institutions can leverage these models to enhance the efficiency of filing Suspicious Activity Reports (SARs).

The present work, consistent with data used by Tiwari et al. (2021) and Tiwari et al. (2022), uses data on 208 limited shell companies incorporated in the United Kingdom (UK), with the names of these entities provided by Transparency International, UK (TIUK), an anti-corruption non-governmental organization (NGO) (Cowdock, 2017). Similarly, to develop and evaluate the detection model, a matching sample of legitimate limited companies from the UK Companies House was used. While it is possible that entities in the matching sample may be involved in illicit activities, consistent with the assumptions of Ravenda et al. (2015), this likelihood is assumed to be minimal.

The data on these entities were obtained from OpenCorporates through a data cleansing and wrangling tool called OpenRefine, using a process outlined by Tiwari (2021) and Tiwari et al. (2021). After cleaning and preprocessing, the data was imported to Neo4J for analysis using graph algorithms, mainly related to determining similarities, communities, and node importance. This approach emphasized the interaction between network structure and node attributes rather than just considering the importance of nodes in a network for analysis (Backstrom and Leskovec, 2011; Seifoddini and Hsu, 1994). The resulting graph scores were exported to Salford Predictive Modeler, a non-parametric tree-based software, where three classification algorithms, namely, Decision Trees, TreeNet, and Random Forests, achieved classification accuracies between 88.17 % and 97.85 %.

The present work highlights the growing interest in the use of data-driven software tools to increase the detection of money laundering activities (Tiwari et al., 2022; Singh and Best, 2016). In light of questions around the effectiveness of current anti-money laundering efforts (Pol, 2018), evolving regulatory landscape, and ongoing money laundering violations, advancing technology and analytics capabilities is critical to reducing false positives and facilitating investigators to focus on high-risk activities (Ray, 2015; Newman, 2007).

The rest of the paper is organized as follows: the ensuing section not only offers a brief overview of research pertaining to shell companies, data mining applications and graph analytics but also delineates the novelty of the present work. This is followed by an exposition on the data sources and the research methodology employed as part of this research. Finally, the paper concludes with an analysis of the findings, an articulation of the study's significance, and potential avenues for future research.

2. Relevant literature

Shell companies, also referred to as front companies or anonymous companies, are entities with minimal operations and assets, often lacking a physical presence. They are used by private companies to become publicly listed (Aydogdu et al., 2007). According to Floros and Sapp (2011), shell companies serve various purposes, including securing tax advantages, avoiding legal liability, facilitating collection of income from patents and other such intangible assets, however, majority of these entities aim to be acquired or merge with another firm. In addition to their role in facilitating reverse mergers, shell companies are also employed as a holding company to protect small entrepreneurs from bankruptcy risks. Consequently, the scholarly landscape is replete with studies examining the use of shell companies for reverse mergers, elucidating both their benefits and inherent disadvantages (Aydogdu et al., 2007; Chen et al., 2016; Chen and Soileau, 2014; Floros and Sapp, 2011; Gleason et al., 2005; Lee et al., 2015; Poulsen and Stegemoller, 2008; Semenenko, 2011; Sjostrom, 2008).

In recent times, these entities have become increasingly linked to a wide range of illicit activities such as bribery, corruption, financial statement fraud, tax evasion, terrorist financing, money laundering, sanction evasion and cybercrimes (Findley et al., 2013; Hubbs, 2018). For instance, in 2002, an anonymous shell corporation named "Anglo-Leasing" was used to launder €24 million of a total €30 million contract intended for updating Kenya's passport system. The anonymity inherent in such entities made it difficult to obtain information about the beneficial owners (Findley et al., 2015; Allred et al., 2017). Several other instances illustrate the misuse of shell companies. For instance, China's ZTE leveraged these entities to evade US sanctions, while the SBM Offshore NV, a Dutch-based group, was found to have paid bribes to shell companies owned by government officials (Hubbs, 2018). Within UK, they have been linked to the laundering of proceeds amounting to £ 80 billion (Houlder, 2017). Additionally, reports by International Consortium of Investigative Journalists (ICIJ) have highlighted the intensive use of shell companies by Politically Exposed Persons (PEPs) and sports personalities to conceal wealth in offshore centers (Harding, 2016).

Consequently, both academic and professional literature focusing on shell companies has focused on its illicit uses (Cooley et al., 2018; Cowdock, 2017; Does de Willebois et al., 2011; Findley et al., 2013; Findley et al., 2015; Gup and Beekarry, 2009; Harding, 2016; Hubbs, 2018; Jancsics, 2017; Stack, 2015a,b). Additionally, the literature has also emphasized the critical role that offshore centers and tax havens have assumed in facilitating the proliferation of shell companies (Alstadsæter et al., 2018; Christensen, 2011; Christensen, 2012; Picard and Pieretti, 2011; Sikka and Willmott, 2010; Stack, 2015a,b,c; Zucman et al., 2015). Sikka and Willmott (2010) examined the role of tax havens and shell companies in tax evasion schemes, revealing a practice where entities sold products to trading subsidiaries at below-market prices, allowing these subsidiaries—often shell companies based in tax havens—to resell the products at market prices and retain the profit margin. Similarly, Christensen (2011, 2012) and Harding (2016) highlighted the role of tax havens in enabling shell companies to obscure illicit financial flows and obfuscate the identities of beneficial owner, thus aiding in corruption.

Zucman et al. (2015) posited that shell companies and offshore accounts allow the rich to invest domestically and internationally while maintaining anonymity, thereby enabling tax evasion. Alstadsæter et al. (2018) further elucidated that approximately ten percent of the total world's Gross Domestic Product (GDP) is held in tax havens, with a significant portion managed through shell corporations held in jurisdictions like the British Virgin Islands and

Panama. Similarly, Gilmour (2020) emphasized the role of complex corporate structures and anonymous shell companies, primarily incorporated through offshore financial centers, to facilitate money laundering. Al-Emadi (2021) subsequently shifted the spotlight to the UK, highlighting it as a hub for illicit and corrupt funds. In response to such findings, the UK initiated measures to ensure a public registry of beneficial ownership.

The United Kingdom pioneered transparency in corporate ownership via implementation of Persons with Significant Control (PSC) register, which require disclosure of ultimate owners in the corporate registry. Ho (2017) offered an overview of these regulations and evaluated the implementation of similar regulation in Hong Kong. The primary impetus behind UK's PSC register was to tackle tax evasion, corporate misconduct, and other illicit activities. The focus of these regulations shifted from legal entities to identifying the natural persons who ultimately own, benefit from, and exert control or influence over the assets of a legal entity.

Lee and Palstra (2018) sought to evaluate the effectiveness of the UK's ownership register, highlighting issues pertaining to data quality and compliance. They concluded that while efforts have been made to enhance transparency in addressing issues surrounding shell companies, transparency should be viewed as a means to an end rather than the ultimate solution. Although a globally transparent governance regime is theoretically desirable, implementing such regulations in practice could prove highly challenging and potentially counterproductive. Similarly, Balakina et al. (2017) pointed out that the demand and supply for tax and financial havens exist because of the need for banking secrecy. As a result, establishing a transparent regime in one jurisdiction may inadvertently increase demand for secrecy in others, thereby reducing overall transparency. Much of the existing scholarly discourse on shell companies predominantly centers on regulations related to international standards aiming to achieve transparency and their implementation, and the oversight of corporate service providers (CSPs).

In response, efforts have been made to develop innovative approaches to combat the concealment and laundering of illicit proceeds, rather than solely concentrating on improving the regulations. Researchers have investigated both machine-learning and traditional statistical approaches to detect money laundering (Bidabad, 2017; Chang et al., 2008; Colladon and Remondi, 2017; Deng et al., 2009; Drezewski et al., 2012; Gao and Ye, 2007; Gao, 2009; Gilmour, 2017; Irwin et al., 2012; Ju and Zheng, 2009; Ngai et al., 2011; Perols, 2011; Regan et al., 2017; Savage, 2017; Savage et al., 2016; Turner and Irwin, 2018; Unger et al., 2011; Wang et al., 2007; Zdanowicz, 2004; Zdanowicz, 2009; Zhang et al., 2003).

The growing focus on money laundering is underscored in the work of Ahuja et al. (2023), whose bibliometric analysis delineated a three-decade upward trajectory on the subject. Among these studies, Tiwari et al. (2020) provided a comprehensive review of the money laundering literature and uncovered attempts to detect laundering undertaken through using real estate, international trade, and high-value portable goods. Similarly, Ferwerda et al. (2013), Zdanowicz (2009), and Unger (2013) directed attention towards the detection of laundering undertaken through real estate and trade, while Turner and Irwin (2018) explored opportunities for laundering presented by technological innovations, such as bitcoin, and proposed detection methods. Gilmour (2017) studied using high-value portable commodities to launder funds in the UK and abroad. Bidabad (2017) suggested mechanisms to detect laundering undertaken through using banking transactions. In a similar vein, Rocha-Salazar et al. (2021) advocated for the employment of neural networks and abnormality indicators on customer transaction data to detect money laundering.

However, attempts to identify shell companies being used for illicit activities, such as bribery, corruption, and money laundering remain in their early stages. Luna et al. (2018) made a notable contribution by addressing the lack of accessible real banking transaction data by creating a banking transaction simulator for shell and regular companies. Using anomaly detection techniques, they analysed variations between expected and observed business transaction profiles to flag potential shell companies for further investigation. However, the model focused only on banking transactions and therefore did not use publicly available information, such as the number of directors and their backgrounds and whether transactions were made to tax havens or countries with a weak banking regime. This is further reiterated by Pawde et al. (2018) who acknowledged the lack of availability of datasets in public domain for detection of shell companies. In response, they proposed a novel approach of generating data synthetically encompassing the concept of collusion. Additionally, Aggarwal and Dharni (2020) employed Benford's Law to distinguish between shell companies and non-shell companies using their financial data. However, the authors noted the inherent limitation of Benford's Law: its inability to discern naturally occurring data from managed data.

The innovative application of graph analytics and data visualization techniques has gained momentum as a promising tool for analysing complex networks (Bolton and Hand, 2002; Chang et al., 2008; Didimo et al., 2011; Needham and Hodler, 2019; Singh and Best, 2016). These approaches address biases associated with domain-specific variables by facilitating feature extraction and score analysis. As per Becerra-Fernandez et al. (2000) and Hawking et al. (2005), representing and analyzing complex networks using graphs would help investigators identify relevant activities. Bangcharoensap et al. (2015) supported using graph algorithm techniques such as PageRank, community detection, and centrality algorithms in developing detection systems.

Monge and Elkan (1997) proposed a find/union algorithm to detect near-duplicates database records. Becerra-Fernandez et al. (2000) developed a system for assisting an analyst in discovering complex networks of potentially illegal activities by correlating through graph visualizations. Krebs (2002) used degree centrality and closeness centrality algorithms to map networks of terrorist cells. Similarly, Morselli and Roy (2008) used the betweenness centrality algorithm to determine brokers' roles in maintaining flexibility in criminal networks. Chang et al. (2008) developed a system for visual analysis of financial transactions through the combination of visual and textual approaches.

Vitali et al. (2011) used the strongly connected component (SCC) algorithm to identify the international ownership structure and the percentage of control held by each owner. The extent of network connectedness indicates the level of intricacy in a network. Didimo et al. (2011) used clustering techniques to design a system for discovering criminal patterns of money laundering and fraud. Kido et al. (2016) proposed applying the Louvain method for modeling topics based on detecting communities in graphs. Identifying communities where members interact more with each other can be challenging. The ability to find and analyze communities can provide more knowledge about the network's structure. Liu et al. (2016) developed graph analysis techniques to apply real-world healthcare datasets to detect fraud, waste, and abuse activities. Representing healthcare relationships through heterogeneous graphs leads to identifying relationships, anomalous individuals, and communities through analysis of local and global graph characteristics. It led to a reduction in the list of suspects and increased focus on genuinely suspicious individuals or activity.

Similarly, motivated by the need to examine money laundering as a process crime, Akartuna et al. (2024) undertook an analysis by mapping numerous combinations of money laundering actions and schemes. These combinations, obtained from a review of money laundering typologies report (Akartuna et al., 2024), were visualized as a network to draw policy-relevant insights for prevention. The authors conducted network-based analyses, specifically, (1) centrality analysis to identify the most critical money laundering actions that should be prioritized for prevention, (2) resilience analysis to facilitate iterative interventions against money laundering actions and assess their impact on the wider network and (3) subgroup analysis to identify clusters of commonly used money laundering schemes or typologies. The findings shed light on the broader dynamics of money laundering and underscore its highly resilient nature to outright prevention owing to substantial crime displacement opportunities. Moreover, it outlines the need to consider money laundering-related prevention measures from a tactical, rather than an outrightly disruptive, perspective. This work emphasizes on the utility of network analysis for examining complex criminal phenomena, including money laundering.

Although there is a breadth of existing research on illicit activities, there does not exist a prior study that develops models to detect illicit shell companies using publicly available information as done in this research; this finding is supported by a prior review by Tiwari et al. (2020). A close exception to this is the work of Rocha-Salazar et al. (2022) who developed a methodology for detecting suspicious patterns in financial networks, with a focus on shell companies. They critically evaluated past works, such as those Joaristi et al. (2019) and Tiwari et al. (2021) for their reliance on centrality metrics. They also critiqued Aggarwal and Dharni (2020) for their reliance on metrics obtained from financial statements.

In contrast, this current work harness data on entities from a vast range of sources, including UK Companies House, OpenCorporates, OpenSanctions, and the Financial Secrecy Index. The comprehensive range of data sources and the information considered, which includes information pertaining to number of directors and their backgrounds, addresses the criticism of relying solely on financial statement metrics. Furthermore, the current research uses a combination of graph algorithms, including centrality, community detection, and similarity, as well as supervised learning to develop detection models, thus addressing the criticism of solely relying on centrality metrics. Overall, the approach presented in the current research addresses the criticisms related to reliance on specific metrics and the need to consider a broader range of information sources which are publicly available.

3. Data

TIUK, an anti-corruption NGO, has kindly supported this project. It has provided a list of cases, names, and company numbers of 804 companies identified as worthy of analysis as shown in Table 1. To come up with a list of these companies, TIUK used open-source data from various investigations dating back to 2004 and records and leaked documents from Open Corporates and UK Companies House, the British corporate registry. It also used this data in a 2017 report called *Hiding in Plain Sight: How UK Companies are used to Launder Corrupt Wealth* (Cowdock, 2017). As mentioned above, money laundering was the central theme for all of them. This TIUK dataset has also been used in prior research by Tiwari et al. (2021); Tiwari (2021).

For this paper, the limited company type was considered due to the availability of information. Among the 210 limited companies, two were found to be duplicates, resulting in 208 companies; the names of the cases for 13 entities were unidentified in the public domain and were categorized as "Unidentified Cases”. Additionally, a matching sample of 205 limited companies (reduced down from 208 because of 3 duplicates) was selected based on their cash balance; when this was unavailable, the period of operations and company status (active, dissolved or in liquidation). Companies from the matching sample were labeled as non-corrupt. The entities in the matching sample may be involved in illicit activities, but this risk is considered very low; consistent with assumptions made by Ravenda et al. (2015), it is assumed that a low probability of illicit activity exists for an entity being chosen from the large population of companies registered with the corporate registry, such as the UK Companies House in this scenario.

Table 1
Data on shell companies involved in illicit activities.

Number of companies	804
Incorporation country	UK
Estimated amount of funds laundered	£ 80 billion
Company types:	436
Limited Liability Partnership (LLP)	158
Limited Partnership (LP)	210
Limited Company (Ltd)	

Information about the final sample of 413 companies (208 corrupt and 205 non-corrupt) was initially collected from OpenCorporates. OpenCorporates has been preferable for investigating and analyzing company data because of its ability to provide what is referred to as "White-Box Data" – which is well-defined, transparent, regularly updated, and accompanied by provenance (OpenCorporates, 2019a,b). Information about the appointment of executives was obtained from the UK Companies House. Other information sources used were EveryPolitician, OpenSanctions, the UK Companies House Disqualified Directors Dataset, and the Financial Secrecy Index.

The names of the companies, along with their company numbers and the corresponding cases, were imported to an open-source piece of software called OpenRefine, facilitating the analysis, cleansing, and transforming of data from one form to another through the use of expression scripting language, also known as General Refine Expression Language (GREL) (Kusumasari and Fitria, 2016; Carlson and Seely, 2017; Hill, 2016). The list of companies was reconciled with OpenCorporates through the OpenCorporates API endpoint at OpenRefine. On matching the company names with target entities using company numbers as identifiers, company data were obtained in "javascript object notation" (JSON) format. The data obtained in JSON format were parsed using GREL to get information for each row item.

The information about the number of appointments of executives in companies was obtained from the UK Companies House. The data obtained were added to the spreadsheet in OpenRefine containing company data. Additionally, web scrapping of the following sources was undertaken:

1. EveryPolitician: A list of all identified politicians was obtained and imported to OpenRefine.
2. OpenSanctions: A list of all individuals and entities with any sanctions imposed on them was obtained and imported to OpenRefine.
3. UK Companies House Disqualified Directors Dataset: A list of all the executives disqualified from acting as directors in UK-incorporated entities was obtained and imported to OpenRefine.
4. Financial Secrecy Index (FSI): The data on the ranking of jurisdictions based on financial secrecy were obtained and imported to OpenRefine.

The information obtained and imported to OpenRefine was used to reconcile the list of executives for the entities with individuals in the lists as mentioned above to extend the data to incorporate any additional information. The ranking of jurisdictions as per the FSI was also included. The dataset and the information on entities is available upon request subject to prior permission from TIUK.

4. Graph construction

Graph analytics offers an efficient approach to handle large volumes of data (Liu et al., 2016). Additionally, visualizing dynamic and complex datasets in the form of graphs provides users with a detailed overview, facilitating them to filter, select, and examine network details. However, the topology of graphs developed from real-world scenarios is heterogeneous, resulting in challenges such as the dense node problem; that is, a small fraction of nodes has connections to a significant fraction of other nodes. Secondly, the graph problems are data-driven, indicating that the computation depends on the graph topology. Consequently, the optimization of the execution of graph algorithms is not possible as it is difficult to determine the computation structure in advance. Thirdly, graph access patterns have poor spatial memory locality, resulting in large amounts of random memory access. Finally, the run time of most graph algorithms is dominated by memory access. Therefore, a graph storage and retrieval technology is required to support a graph data model that is productive, capable of storing large graphs, and can execute complex queries on large graphs. Neo4J is capable of addressing all these challenges (Cattuto et al., 2013).

Graph database technology enables the optimization of interrelated and densely connected datasets to allow for the construction of predictive models and the identification of patterns through the use of traversal queries (Miller, 2013). It uses graph models to store, manage, and update relationships and data with free and flexible schemas (Huang and Dong, 2013). Neo4J, a graph database platform, was chosen to develop graphs from the data collected above. It involves using a query language called 'Cypher' and lends flexibility in setting up the data model. It provides features such as online transaction processing on graph data (OLTP) and graph analytics. Similar to path algebras, graph query languages such as 'Cypher', with different levels of expressivity, serve as a function for querying graphs for data. 'Cypher' facilitates the application of algorithms to complex network data (Aleta and Moreno, 2019; Rodriguez and Shinavier, 2010; Memon and Wiil, 2014).

The flexibility in setting up the data structure exposes hidden relationships in the data and allows conclusions to be drawn accordingly (Van Bruggen, 2014). In Neo4J, the graph data model comprises the following building blocks:

- Node: A term used to refer to the entities in the data.
- Relationships: Establish the connections between nodes and thus provide a way of structuring the data.
- Properties: Fundamental attributes describing both the nodes and relationships.
- Labels: A naming convention for nodes and relationships facilitating the creation of subgraphs, easing investigation and determination of hidden data patterns.

Fig. 1 below provides an overview of the overall approach utilized. The data collected through the procedure above are imported as two CSV files. The data collected on 413 limited liability companies were randomly separated into training and testing samples of 320 (77.481 %) and 93 companies (22.518 %), respectively. The training sample data were then imported to Neo4J using 'Cypher', while keeping the testing sample separate for evaluating model performance on holdout data. The set of queries used to import and establish the database structure is available from the corresponding author upon request.

A graph comprises of nodes and relationships, and this aspect can be effectively used to model highly interconnected data. As part of this paper, nodes for this graph database model are cases, companies, its owners and executives, addresses, and previous company names. The properties describe the attributes of nodes, and the relationships indicate the existing links between the nodes. Using properties and labels along with nodes and relationships in Neo4J allows taking advantage of property graphs. As per Miller (2013), property graphs are attributed, labeled and directed multi-graphs, facilitating the representation of the most complex data in the form of graphs. The graph database schema used in this research is provided in Fig. 2 below.

The root node is listed as Cases, and it has an immediate connection with the Company node. For the matching sample, the cases are labeled "Non-Corrupt" followed by a number to ensure the data schema remains consistent and the problem of node density can be avoided by attributing a numerical value to each company. A company is registered at an address which is denoted by "Company Address" in the database.

Additionally, a company will have executives (denoted by "Executive") and owners (represented by "Ultimate Owner"). Finally, a company may have a previous name (indicated by "Previous Company") along with a previous address (denoted by "Address"). In a scenario where a company has a previous address and an earlier name, the relationship would be depicted by "Change in Address" relationship to the "Company" node, while the "PreviousName" relationship would connect to the "Previous Company" at that earlier address.

The network construction is inspired by the views of Fakhraei et al. (2015), who state that models incorporating the multi-relational nature of the networks gain predictive performance. This also provides an opportunity to run some graph algorithms on the dataset and draw insights which could lay the foundation for future research. It is consistent with the views of Drakopoulos et al. (2015). They state that graph databases not only act as a means for graph storage but offer opportunities to perform graph analytics techniques. Graph algorithms related to determining similarities, communities and node importance can be applied. One of the underlying rationales is to establish a base for the interaction between network structure and node attributes rather than just considering the importance of nodes in a network as it takes advantage of only network structure and not node attributes (Backstrom and Leskovec, 2011).

While analyzing networks, a wide range of centrality measures and clustering algorithms is possible. The lack of literature on which methods might yield better results in a wide variety being evaluated is acknowledged in the present work.

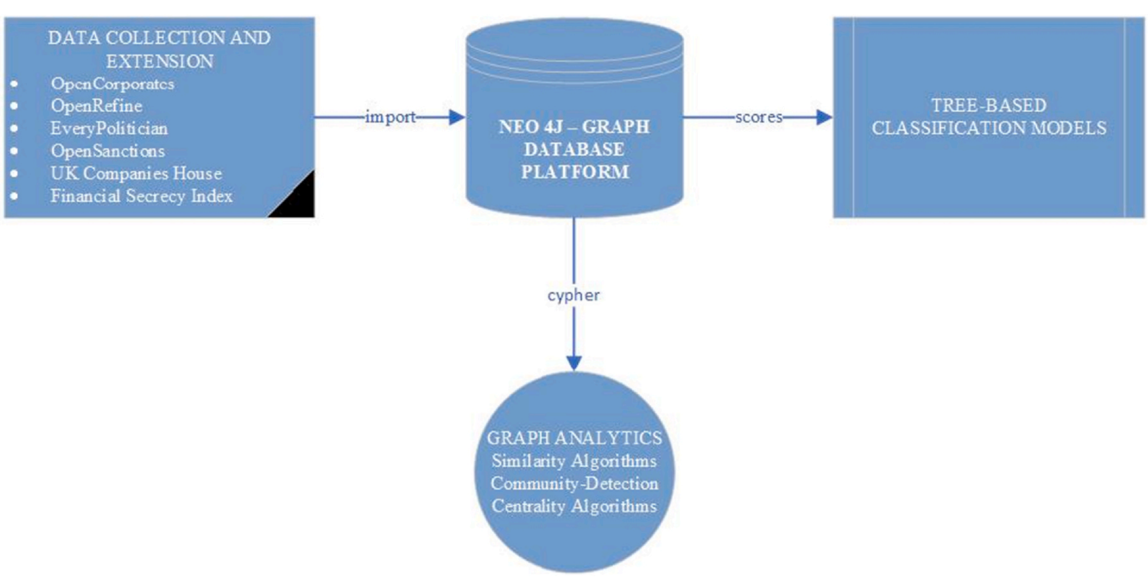

Fig. 1. Overall methodology, which is consistent with prior research (Tiwari, 2021; Tiwari et al., 2022; Tiwari et al., 2021).

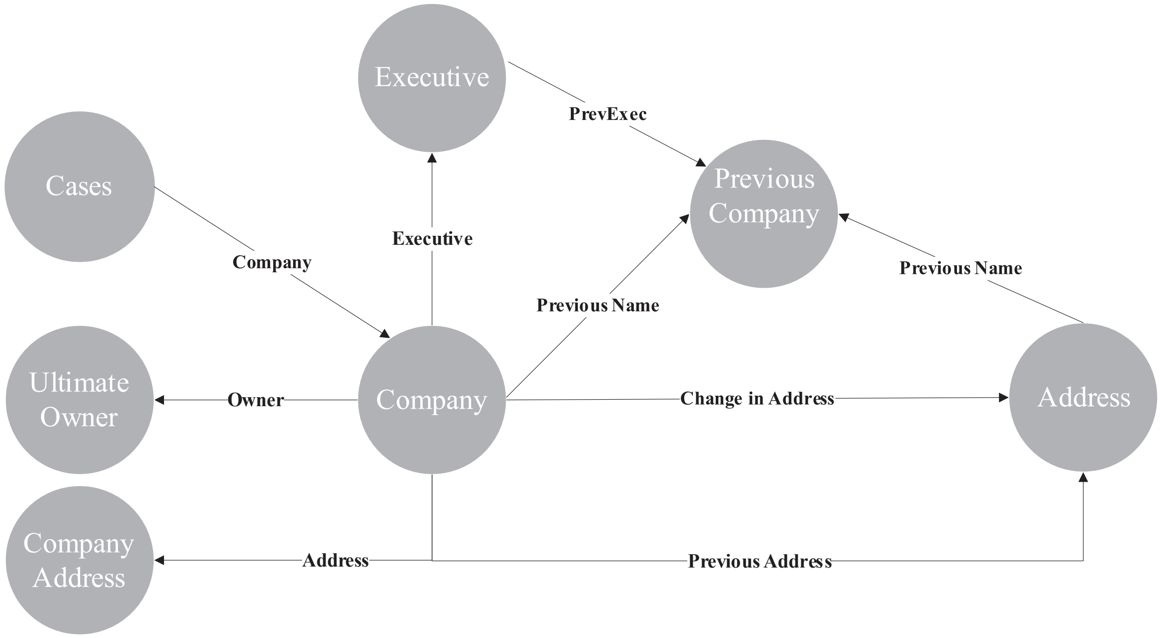

Fig. 2. Database schema for the data model, which is consistent with prior research (Tiwari, 2021a; Tiwari et al., 2022; Tiwari et al., 2021b).

5. Methodology

The present work employs a quantitative methodology, employing graph analytics and supervised learning techniques to evaluate the structural and relational attributes of entities involved in a money laundering network, consistent with the approach adopted by Tiwari et al. (2022) in their investigation of a financial crime network.

The paper applies algorithms aimed at identifying similarity and community among nodes as well as the centrality of nodes in the network to come up with scores to be used for a classification model (the

Table 2
Overview of the algorithm categories used.

Algorithm category	Overview	Motivation, objective or application
Similarity	• Similarity-metric based approaches compute the similarity of node pairs as per the attributes of a node or topological structures.	• Limitations of feature selection, unbalanced output classes and problems of computational cost make similarity-metric based methods attractive (Yao et al., 2016; Yao et al., 2018). • These approaches are simple and have exhibited superior performance on complex networks and outperformed machine-learning approaches in the past (Lu et al., 2017).
Community – Detection	• A community in a network can be categorized as a group of objects which are similar to each other and dissimilar from the remaining objects in the network. • The identification of communities could aid in inferring similar behavior displayed by a group, discovering nested relationships and preparing the data for use in further analysis (Gao and Ye, 2007; Kido et al., 2016; Needham and Hodler, 2019).	• A similarity measure is computed between objects, and a clustering algorithm is used to group those objects into families based on a corresponding similarity measure (Seifoddini and Hsu, 1994). • The underlying principle is that similar nodes exhibit dense interconnectedness among themselves and sparse connectivity to objects in the network (Needham and Hodler, 2019; Raghavan et al., 2007). • It also has an application in identifying functional modules in biochemical networks (Raghavan et al., 2007). • Luna et al. (2018) used companies with the same bank accounts to classify communities of shell companies in their study. • Gao and Ye (2007) analyzed money laundering networks through link analysis (comprising manual, graphical and structural approaches), community generation (use of techniques such as K means and clustering to generate binary relationship-based communities), and network destabilization (comprising power analysis, cohesion, and role analysis) to disturb the flow in networks.
Centrality	• Centrality algorithms facilitate understanding the nodes' importance and group dynamics (Needham and Hodler, 2019).	• The rationale for using centrality algorithms lies in a phenomenon known as homophily in network science. Homophily refers to the tendency that connections exist between fraudulent entities and these entities work in collusion to accomplish an objective (Bangcharoensap et al., 2015). • The most commonly used measures are Degree Centrality, Closeness Centrality, and Betweenness Centrality, as proposed by Freeman (1978). As per Ruhnau (2000), all three measures depend on the size of the graph. • As per Chen et al. (2012), Betweenness Centrality and Closeness Centrality are prominent geodesic-path based ranking measures. • A well-known variation of the Eigenvector centrality is the PageRank algorithm proposed by Page et al. (1999).

codes used for running the algorithms are available upon request). More specifically, the Graph Data Science Playground App within the Neo4J environment was used to examine similarities between nodes using the Jaccard or Overlap similarity metrics or both. Consequently, the similarities obtained were used to establish relationship between nodes in the network. Subsequently, the Graph Data Science Playground App was utilized to run algorithms related to community detection and centrality. This approach of using a combination of similarity, centrality, and community detection algorithms was applied consistently to both the training and testing samples. The academic literature supports the use of community detection approaches in tandem with other graph algorithms to yield better results (Kido et al., 2016; Needham and Hodler, 2019; Raghavan et al., 2007; Yao et al., 2018; Zuo et al., 2012). Finally, the approach adopted in the present work aligns with the views of Hao et al. (2018), who argue that such an approach allows accounting for the identification of overlapping communities, which can aid in determining node influence, as the number of communities to which a node is representative of its propagation capacity.

Table 2 presents the overall graph algorithm categories considered, before the specific graph algorithms used in this research are presented in Table 3. Various combinations of similarity, community detection and centrality algorithms are utilized; namely, Jaccard Index, Overlap Index, Triangles and Triangle Counting Algorithm, Clustering Coefficient, Betweenness Centrality, Approximate Betweenness Centrality, and Harmonic Centrality algorithms.

This process of using the Graph Data Science Playground App to run similarity, centrality, and community detection algorithms was applied consistently to both the training and testing samples. The resulting scores obtained in Neo4J from the graph algorithms specified in Table 3 were retrieved and exported to Salford Predictive Modeler to develop supervised learning classification models that try to predict the binary classification of corrupt (denoted by 1) or non-corrupt (denoted by 0).

For this paper, three supervised learning techniques were used, namely CART decision trees, TreeNet and Random Forest models. All three of these models are non-parametric, which is necessary in this research because there is no prior literature on this topic and so it is not possible to propose a specific parametric model structure *a priori.* Furthermore, all three techniques are tree-based which means they are immune to outliers and can model any interactions between the different algorithms used, which importantly means that multi-collinearity is not an issue.

The use of tree-based models also has support in prior fraud detection literature as well as other areas such as medicine, engineering, finance and marketing (Gepp et al., 2010; Kumar and Bhattacharya, 2006; Kumar and Haynes, 2003). Link analysis and graph mining have been used in law enforcement, anti-terrorism, and other security areas (Phua et al., 2010). Al Hasan et al. (2006), in their study of the coauthorship network, laid the foundation for extraction of features to be used in supervised learning for a systematic link prediction. They considered factors such as proximity and individual attributes to extract features from the network. They found that considering link prediction as a binary classification problem and using useful features in a supervised learning framework improved accuracy. On using the PageRank algorithm to rank features, Ienco et al. (Ienco et al., 2008) found that higher feature-ranking led to the influence of a node in a network which resulted in increased classification accuracy.

All models were first built on training data, before they were evaluated on the holdout/testing test data to determine how they would perform on previously unseen data, which is akin to future performance.

6. Results

Tables 4–6 present the performance of the detection models using a variety of combination of scores obtained from graph algorithms and supervised learning techniques, including Decision Trees, TreeNet, and Random Forests. The performance of each combination was evaluated using multiple performance metrics consistent with prior literature (Kute et al., 2021) and best practice in data analytics; specifically, area under receiver operating characteristic curve (AUROC), precision, recall and F-value. The detailed outputs generated using the aforementioned classification models, based on various combinations of graph algorithm scores, are presented in the appendices.

Table 4, as presented below, outlines the performance of Decision Tree models across various combinations of algorithms. The combination using Jaccard Similarity metrics to establish relationship between nodes, and then utilizing Triangle Counting, Clustering Coefficient, Betweenness, Harmonic, and PageRank algorithms achieved the highest accuracy of 97.85 %, exhibiting exception effectiveness in detecting illicit shell companies. This is further supported by AUROC of 97.79 %, reflecting strong discriminatory power. The high values of precision and recall (97.67 % for both) suggest a balanced performance with minimal false positives. In contrast, the combination using Overlap Similarity to establish relationship between nodes, and then using Triangle Counting, Clustering Coefficient, Approx. Betweenness, Harmonic, and Eigenvector centrality exhibited a lower accuracy of 88.17 %, suggesting a need for improvements for higher accuracy. Despite this, the AUROC of 92.12 % suggests that the model retains some discriminatory capability.

TreeNet models, as summarized in Table 5 below, demonstrated slightly superior performance compared to Decision Trees. The combination using Jaccard Similarity metrics to establish relationship between nodes, and then utilizing Triangle Counting, Clustering Coefficient, Betweenness, Harmonic, and PageRank algorithms, again emerged as the best-performing, achieving an accuracy of 97.85 % and an AUROC of 98.14 %, the highest among all models. Furthermore, precision and recall values remained consistent at 97.67 %, thus reaffirming the model's ability to detect illicit shell companies while minimizing false positives. Interestingly, substituting Betweenness with Approximate Betweenness Centrality algorithm resulted in slight decrease in accuracy (93.55 %).

Random Forest models, as demonstrated in Table 6 below, performed competitively, especially with a combination of Jaccard Similarity, Triangle Counting, Clustering Coefficient, Betweenness, Harmonic, and Eigenvector centrality algorithms, achieving an accuracy of 97.85 % and a recall value of 95.35 %. The AUROC of 97.30 % suggests that the model performs well in distinguishing between corrupt and non-corrupt entities. In contrast, the combination involving Overlap Similarity, Triangle Counting, Approx. Betweenness, Harmonic and Eigenvector centrality algorithms achieved a reduced accuracy of 91.50 %. Despite this, Random Forest models exhibited robustness across multiple configurations, maintaining relatively high precision and recall values.

Overall, consistent performance is exhibited across most models with classifications accuracies ranging up to 97.75 % on using different combination of graph algorithms. This sets a solid benchmark for future comparative research. The lack of any work in the academic literature on the detection of shell companies using publicly available information prohibits comparison to evaluate the performance of the present methodology. However, this work has now set a benchmark for future studies to compare against.

7. Discussion and conclusion

While there has been growing attention to the identification of economic networks in trading, products and detecting fraudulent companies (Pacini et al., 2016; Vitali et al., 2011; Rocha-Salazar et al., 2022), to our knowledge, no prior quantitative analysis has previously focused on illicit shell companies using publicly available information. Furthermore, our approach also addresses the criticisms associated with reliance on specific metrics; this is in line with the works of Mondragon

Table 3
The specific graph algorithms used.

Specific Algorithm –Category	Overview	Brief Description and Application
Jaccard Index – Similarity	• Paul Jaccard proposed the Jaccard Index in 1901 (Lü and Zhou, 2011). The Jaccard index is used for information retrieval and is a normalized version of the Common Neighbor similarity algorithm by considering the influence of a node in the network.	• It solves the problem of the Common Neighbor similarity algorithm by emphasizing the links of non-influential nodes to make sure that the common neighbors among them exist because of their similarity and not influence (Lu et al., 2017). • It measures similarity as a proportion of elements two objects have in common in comparison to elements that are not common between them. It is distinct from an index such as Cosine similarity which considers information related to vector similarity (Lewis et al., 2006). • The value of the Jaccard Index varies between the range of 0 and 1 with a value close to 1 indicating a higher proportion of similarity; and can be used for community detection especially in the case of dense graphs (Needham and Hodler, 2019; Raghavan et al., 2007).
Overlap – Similarity	• The Overlap Similarity algorithm, also known as the Hub Promoted Index (HPI), aims at measuring the overlap between two sets (Lü and Zhou, 2011).	• It considers the strength that weak connections share among themselves (Granovetter, 1973).
Triangles (measured by Clustering Coefficient) and Triangle Count Algorithm – Community Detection	• In a graph network, a triangle refers to a set of three nodes whereby each node is related to all other nodes. The triangle counting algorithm aims at detecting communities by determining the number of triangles passing through each node in the graph network (Needham and Hodler, 2019).	• The technique gained immense popularity in network analysis because of the size of networks such as the world wide web (Schank and Wagner, 2005). • Van Vlasselaer et al. (2017) aimed at identifying companies intentionally going bankrupt to avoid tax payments. They did so by assessing the effectiveness of network information in social security fraud detection using network-based features such as triangles to identify such companies. • The importance of triangles is quantified by the clustering coefficient which is interpreted as curvature (Eckmann and Moses, 2002).
Betweenness – Centrality	• Betweenness Centrality, as proposed by Freeman (1977, 1978), determines the influence a node has over the flow of information in a graph. It tends to find nodes that act as a bridge in a graph network between other nodes.	• Morselli and Roy (2008) examined the importance of brokers in criminal networks using the betweenness centrality algorithm. They found the position of a broker critical in resource pooling and coordination of activities.
Approximate Betweenness – Centrality	• Randomized – Approximate Brandes ("RA-Brandes") algorithm, also known as approx. betweenness, enables fast calculation of approximate scores of betweenness centrality between nodes of a graph (Needham and Hodler, 2019).	• It involves the calculation of shortest paths between a subset of nodes instead of all pairs of nodes in the graph. The nodes may be selected uniformly at random, or there may be random selection of nodes with high degrees.
Eigenvector – Centrality	• Eigenvector Centrality was pioneered by the works of Katz (1953) and Bonacich (1972, 1987).	• It is based on the idea that the node is central if it is related to central nodes. The value of centrality instead of the number of adjacent nodes can be the determining criteria.
PageRank – Centrality	• PageRank Centrality algorithm, a variant of Eigenvector centrality, measures the connectivity of nodes (Needham and Hodler, 2019). • It involves traversing through a graph and computing the frequency of hitting each node during these traversals. • The algorithm provides a network centrality score in light of the entire graph structure.	• At times, the goal is to determine the importance relative to a small subset of the objects (Gleich, 2015). • PageRank methods have been used to study engineered systems such as road and urban space networks (Jiang et al., 2008). • Zuo et al. (2012) used PageRank combined with community detection and known brain regions to understand the changes in brain structure across a population of 1000 people who correlate with age. • Mooney et al. (2012) used it to determine changes in the network of molecules linked by hydrogen bonds among water molecules.
Harmonic – Centrality	• The Closeness Centrality algorithm aids in node detection in a graph that aims to spread information through a subgraph. However, the performance of the algorithm is subject to the connectivity of nodes in a graph network (Needham and Hodler, 2019). • A variant of closeness centrality, to address the issue of real-life networks of the absence of connectedness, is harmonic centrality, also referred to as valued centrality (Needham and Hodler, 2019).	• Krebs (2002) used closeness centrality scores to reveal terrorists in network cells vital for carrying out their operations. Colladon and Remondi (2017) used the closeness centrality measure to identify factoring businesses in Italy responsible for undertaking trade-based money laundering activities.

et al. (2018), Malinick et al. (2013), and Hite et al. (2005) who note the need to consider a broader range of information sources. This paper also capitalizes on the suggestion made by Baker and Faulkner (1993), calling for archival data to outline relationships among entities of a network. By using the networks prevalent among entities in a corrupt network and analyzing the links and similarities, the present work contributes to the field of financial crime detection by developing and evaluating models for detecting illicit shell companies via integration of graph-based metrics and supervised learning techniques, thus attempting to address the costly problem of money laundering annually valued in excess of one trillion dollars.

Motivated by the works of Barabási (2009), who emphasized the existence of commonality in networks, and development of detection systems for fraudulent activities (Bangcharoensap et al., 2015; Battaglia et al., 2018; Van Vlasselaer et al., 2017), the findings of the present work highlight the importance of analyzing complex network structures

Table 4
Performance of decision tree with different combinations of algorithms.

Combination of algorithm scores	Accuracy	AUROC	Precision	Recall	F-value
Jaccard, Triangle Counting, Clustering Coefficient, Approx. Betweenness, Harmonic and Eigenvector	91.88 %	95.093 %	97.50 %	90.70 %	93.98 %
Jaccard, Triangle Counting, Clustering Coefficient, Betweenness, Harmonic and PageRank	97.85 %	97.79 %	97.67 %	97.67 %	97.67 %
Jaccard, Triangle Counting, Clustering Coefficient, Betweenness, Harmonic and Eigenvector	92.47 %	94.402 %	97.50 %	89.16 %	91.93 %
Overlap, Triangle Counting, Clustering Coefficient, Approx. Betweenness, Harmonic and Eigenvector	88.17 %	92.116 %	94.44 %	79.07 %	86.08 %
Jaccard, Overlap, Triangle Counting, Clustering Coefficient, Approx. Betweenness, Harmonic and Eigenvector	93.55 %	93.977 %	95.12 %	90.70 %	92.86 %

Table 5
Performance of treenet with different combinations of algorithms.

Combination of algorithm scores	Accuracy	AUROC	Precision	Recall	F-value
Jaccard, Triangle Counting, Clustering Coefficient, Approx. Betweenness, Harmonic and Eigenvector	93.55 %	98.95 %	95.12 %	90.70 %	92.86 %
Jaccard, Triangle Counting, Clustering Coefficient, Betweenness, Harmonic and PageRank	97.85 %	98.140 %	97.67 %	97.67 %	97.67 %
Jaccard, Triangle Counting, Clustering Coefficient, Betweenness, Harmonic and Eigenvector	94.34 %	97.977 %	97.50 %	90.70 %	93.98 %
Overlap, Triangle Counting, Clustering Coefficient, Approx. Betweenness, Harmonic and Eigenvector	94.62 %	95.140 %	97.50 %	90.70 %	93.98 %
Jaccard, Overlap, Triangle Counting, Clustering Coefficient, Approx. Betweenness, Harmonic and Eigenvector	94.34 %	96.442 %	95.12 %	90.70 %	92.86 %

Table 6
Performance of random forests with different combinations of algorithms.

Combination of algorithm scores	Accuracy	AUROC	Precision	Recall	F-value
Jaccard, Triangle Counting, Clustering Coefficient, Approx. Betweenness, Harmonic and Eigenvector	93.55 %	95.558 %	95.12 %	90.70 %	92.86 %
Jaccard, Triangle Counting, Clustering Coefficient, Betweenness, Harmonic and PageRank	96.77 %	97.349 %	97.62 %	95.35 %	96.47 %
Jaccard, Triangle Counting, Clustering Coefficient, Betweenness, Harmonic and Eigenvector	97.85 %	97.302 %	97.62 %	95.35 %	96.47 %
Overlap, Triangle Counting, Clustering Coefficient, Approx. Betweenness, Harmonic and Eigenvector	91.40 %	93.558 %	97.30 %	83.72 %	90.00 %
Jaccard, Overlap, Triangle Counting, Clustering Coefficient, Approx. Betweenness, Harmonic and Eigenvector	93.55 %	96.72 %	95.12 %	90.70 %	92.86 %

to uncover the hidden control and relationships, thus offering a quantitative benchmark for future research and applications. The evaluation of detection models resulted in consistent and high performance when utilizing combinations of graph-based metrics, such as Jaccard Similarity, Triangle Counting, Clustering Coefficient, Betweenness, Harmonic, and PageRank centrality. These combinations provided high level of accuracy (up to 97.85 %) and AUROC values (up to 98.14 %), especially when using TreeNet, which also achieved balanced precision and recall rates of 97.67 %. The use of Random Forest models also exhibited strong performance, achieving a recall value of 95.35 %.

In contrast, models using combinations such as Overlap Similarity and Approx Betweenness exhibited comparatively lower accuracy, directing the need to focus on selection of graph-based metrics. For instance, a Decision Tree employing these metrics achieved an accuracy of 88.17 % and AUROC of 92.12 %, respectively, thus suggesting that while such combinations retain discriminatory capability, further optimization could enhance detection performance. Overall, the present work highlights the role of graph-based metrics and supervised learning models for detecting illicit shell companies.

The key stakeholders to benefit from such models range from legal and compliance professionals to government officials, especially with a tax brief, and anti-corruption NGOs. In addition, auditors, regulatory agencies, and banks with access to transaction information could use such models alongside suspicious transaction analysis to increase detection rates. The developed model would assess the risk of a company being involved in illicit activities and whether further investigation is needed.

Moreover, the results of the present work underscore the need for corporate transparency and effective regulation, as outlined by Cowdock and Simeone (2019). Jurisdictions such as the UK and Singapore have been responsible for supplying opaque offshore structures, thereby reducing the effectiveness of the efforts of transparency and integrity. For instance, the work of Gilmour (2020) and Al-Emadi (2021) have highlighted the role of UK as a destination for illicit funds due to its position in the global financial markets. Despite the efforts made, the UK still faces risks related to money laundering due to lack of beneficial ownership transparency leading to concealment of corrupt wealth which in turn hinders efforts to trace illicit finance. Consequently, there has been a need on their behalf to take additional responsibility (Lasslett, 2019).

8. Limitations and future directions

In recent years, as noted by Nougayrède (2019), efforts to deter the illicit use of shell companies have focused on accelerating intergovernmental measures to prevent tax evasion, money laundering and corruption, such as the cross-border exchange of financial information. A major focus of these measures has been at promoting corporate transparency, such as in the UK, driven partly by the documented relationship between transparency and financial performance. For instance, Akhigbe et al. (2017) found that increased transparency positively impacts banks' financial performance. Despite evidence of public and private actors' non-compliance and failure to implement regulations and recommended standards, regulatory authorities and law-enforcement agencies have advocated for greater transparency in beneficial ownership, on account of the widely accepted notion that greater transparency would facilitate prevention and detection of unlawful activities. While the full impact of these efforts is yet to be ascertained, their success will be dependent on the proactiveness of national regulators and the information quality maintained by local service agents. This ongoing need for better detection mechanisms underscore the importance of developing robust systems, like the models proposed in the present work, to effectively detect and address the misuse of shell companies.

This study serves as a starting point in an essential area for future research. In an ever-changing landscape of regulatory framework, implementing effective regulations poses significant challenges. Therefore, it would be interesting to determine whether an alternative approach, when integrated with existing frameworks, could enhance efforts to combat the use of shell companies for illicit purposes, thus presenting a compelling avenue for further investigation. However, the present study is not without limitations, which presents opportunities for exploration and refinement in the future.

One limitation stem from the reliance on publicly available data. While these sources enhance transparency and accessibility, they may suffer from errors, inconsistencies, or incomplete records, which could have an influence on the reliability of results. Furthermore, an analysis based on publicly available data may lack the depth and granularity offered by transactional data. Therefore, incorporating information, such as transactional data, in future research could complement publicly available data, enhancing depth and accuracy of analysis.

Furthermore, the absence of weighted relationships in the present work limits the ability to capture significance of interactions between entities. In line with the views of Bangcharoensap et al. (2015), who found weighted degree centrality to be effective in distinguishing between legitimate and fraudulent users, incorporating weighted measures of relationships may enhance model accuracy. Similarly, integrating proximity measures between entities could provide meaningful insights into whether spatial closeness is used to mislead authorities and facilitate illicit activities. The dynamic evolution of relationships in a network can be considered in future research.

Moreover, simulating variations in network topologies and application of weights to relationships could provide a better understanding of metric utility across diverse network structures. Future studies could also employ large language models (LLMs) to examine alternative network topologies systematically. They can be utilized to assess whether variations in network structures impact the effectiveness of detection algorithms, thus expanding the scope of this research.

Additionally, the results obtained as part of this study raise questions that warrant further exploration. For instance, examination of why certain similarity metrics was obtained between nodes in the data could facilitate deeper insights into the underlying data patterns. On similar lines, investigation into why certain algorithm combinations outperformed others would provide valuable insights on how these methods interact and complement each other, thus leading to improved approaches for network analysis.

While the present works employs traditional graph algorithms, exploration of other algorithms for assessment of node influence (Chen et al., 2012; Hao et al., 2018; Liu et al., 2018; Lv et al., 2019; Raychaudhuri et al., 2020; Salavati et al., 2019; Srinivas and Rajendran, 2019; Wu et al., 2019; Zhang, 2014; Fei et al., 2018) could add further value. This is in alignment with the views of Lü et al. (2016) who outlines the need for development of centrality measures for identification of influential nodes. Moreover, establishing performance benchmarks for these measures could aid in their application in future research.

Finally, the findings of the present work may be constrained by its focus on UK-based entities, which could limit the generalizability of the results to jurisdictions with varying regulations. Therefore, integrating the analysis to include data from diverse geographical and legal context could enhance the robustness of the findings.

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CRediT authorship contribution statement

Kuldeep Kumar: Supervision. **Adrian Gepp:** Supervision. **Milind Tiwari:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jeconc.2024.100123.

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